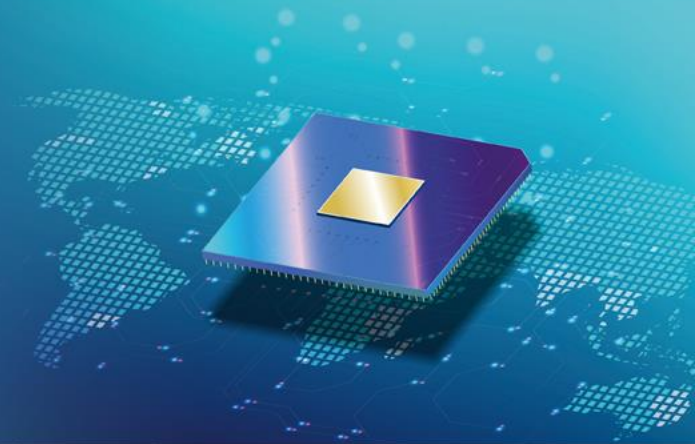


컴퓨터를 활용한 리소그래피 PBL 과제 발표

20211488 선승현 (전자정보공학부 전자공학)
20211486 박재원 (전자정보공학부 전자공학)
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실습 목표

- default/split recipe 설정하여 OPC 후 Mask layout 변화를 확인
- Contour simulation 결과 비교
- Recipe Parameter 변화에 따른 OPC 보정 분석

	default	split	}	Feature Classification
<u>min_feature</u>	65	50		
	default	split	}	Dissection
<u>Dissection.out_vert_len</u>	60	50		
<u>Dissection.in_vert_len</u>	60	50		
<u>Dissection.nominal_segment_len</u>	50	40		
<u>Dissection.min_seg_len</u>	40	35		
<u>Dissection.num_run_segs</u>	5	6		
	default	split	}	Retarget
<u>Retargeting.vert_control</u>	1.0	0.8		

Feature Classification

- Layout 안에 있는 각 Edge가 어떤 종류의 Feature인지 분류하는 단계

→ OPC는 Feature종류(Line/Space/Inside/Jog...)에 따라 보정 방식이 다르기에 분류해야 함

Feature Classification

Design Rule:

<min_feature> 65

<jog_height> 10

Short Edges:

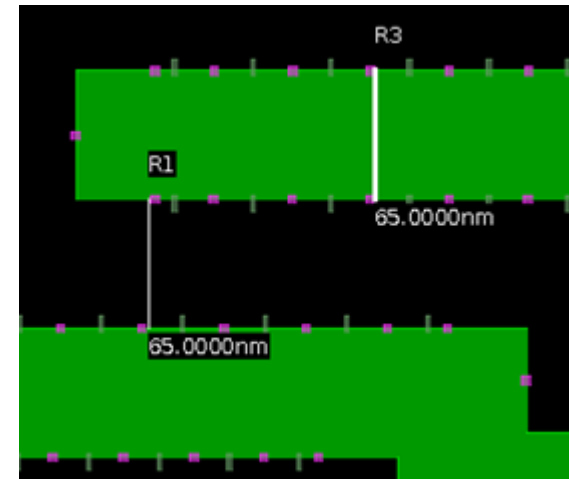
<min_correction_length> {0.5*min_feature}

Line Ends:

<end_width> {2*min_feature}

<min_end_extension> min_correction_length

- Min_feature** : layout의 최소 CD(Critical Dimension)



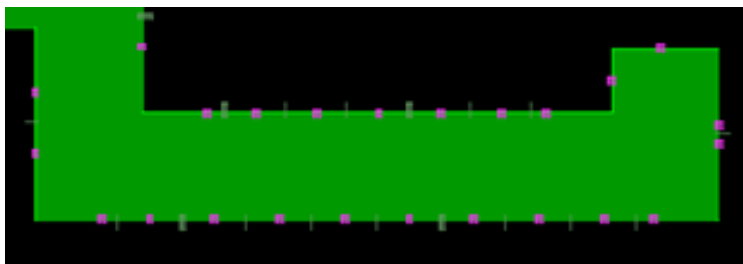
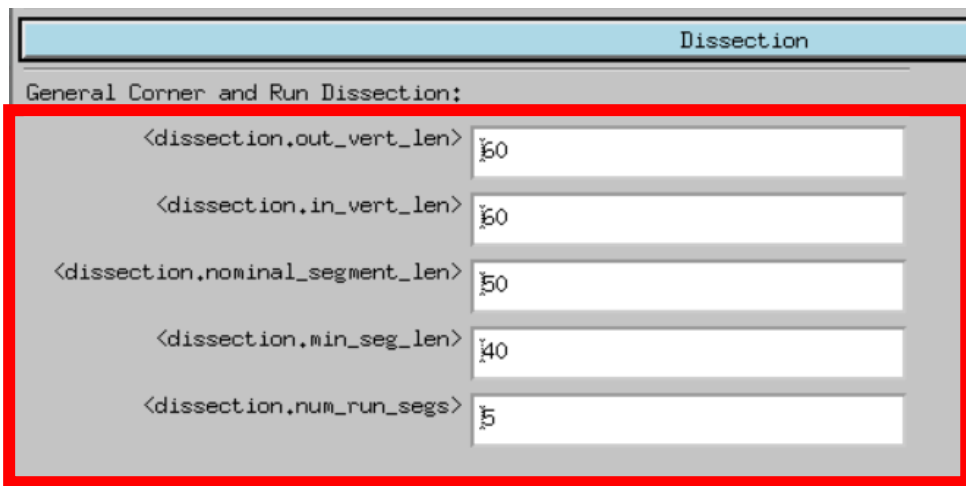
- Jog_height** : Jog로 판단할 최대 길이,
→ 10nm보다 작으면 Jog로 인식

※ Short Edge : OPC 보정을 수행할 최소 길이 / Line Edge : Lien End로 분류할 최대 Edge 길이

Dissection

- OPC 보정을 위해 Edge를 n개의 Segment로 분할하는 과정

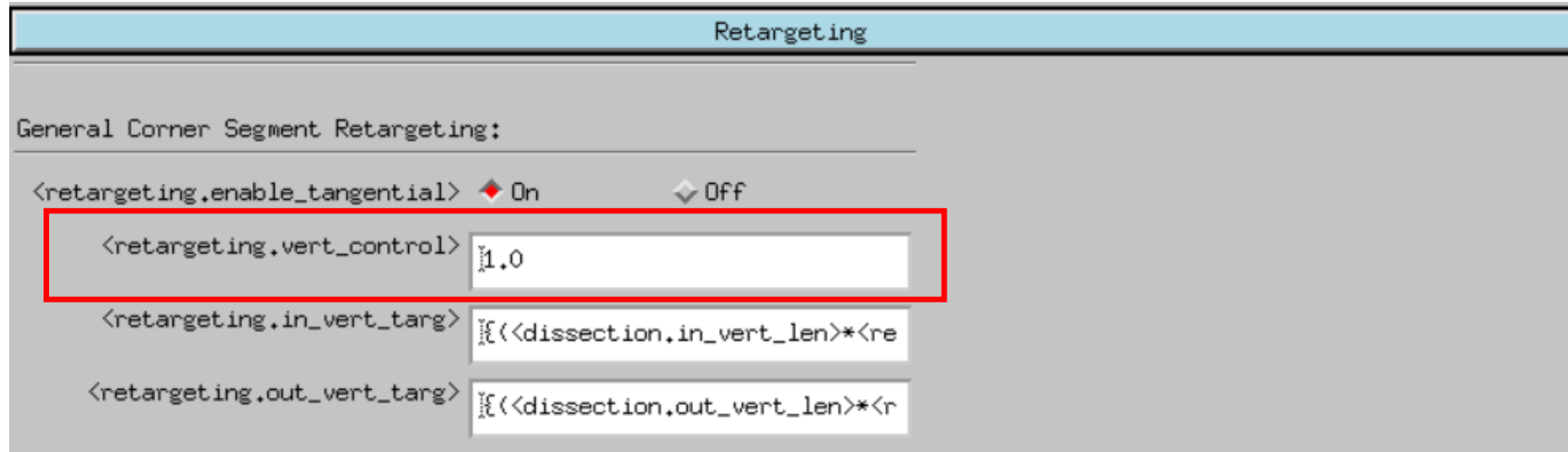
➔ OPC는 Segment단위로 보정량을 계산하므로 Dissection결과가 핵심 과정



- Dissection.out_vert_len
: 외부 코너(Outside Corner) 주변에 생성되는 Segment 길이
- Dissection.in_vert_len
: 내부 코너(Inside Corner) 주변에 생성되는 Segment 길이
- Dissection.normal_segment_len
: 직선 영역(Run segment)의 기준 Segment 길이
- Dissection.min_seg_len
: 분할된 Segment의 최소 허용 길이
- Dissection.num_run_segs
: 직선 영역(Run segment)에서 생성할 Segment 개수
➔ 값이 클수록 Edge가 더 세밀하게 분할됨

Retargeting

- OPC 수행 전 Edge의 Target 위치를 조정하는 단계
- ➔ 공정에 의해 발생하는 패턴 왜곡을 미리 고려하여 보정 목표점을 Retarget하는 것



- Retarget_vert_control : Vertex(코너) 부근의 Target 위치 조정 강도

Hierman setting / Corexec

```
[litho42@ssudisu4 ~/WK13]$ hierman default.pjx
```

hierman

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/user/tools/synopsys/proteus_lx/Q-2020.06-4/linux64/bin/hierman
host: ssudisu4

```
[litho42@ssudisu4 ~/WK13]$ corexec default.pjx
```

corexec

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/user/tools/synopsys/proteus_lx/Q-2020.06-4/linux64/bin/corexec
host: ssudisu4

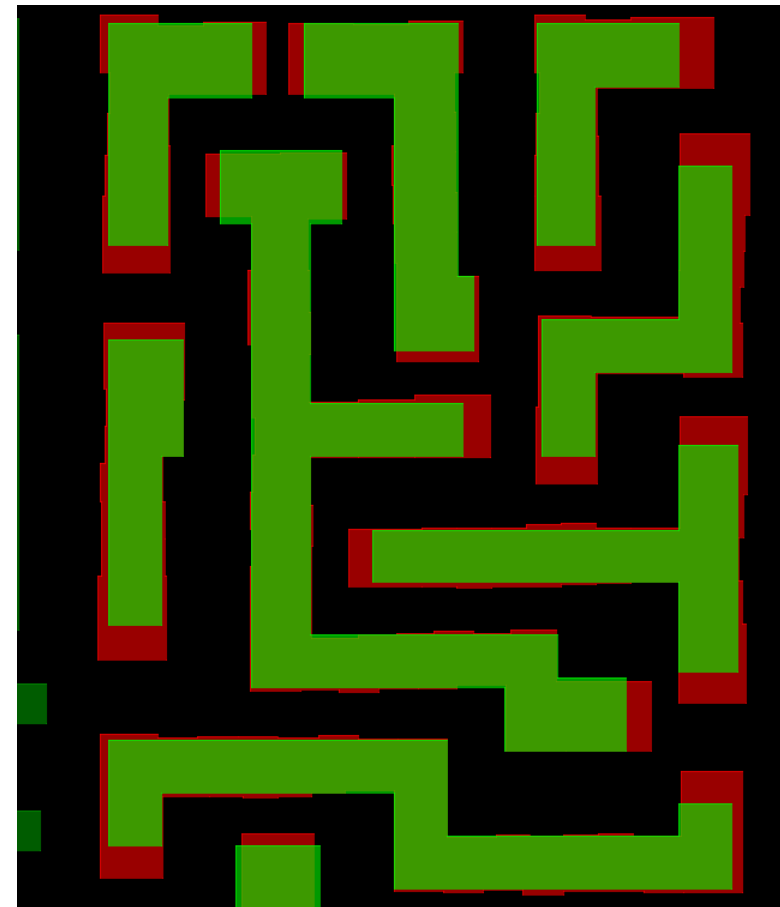
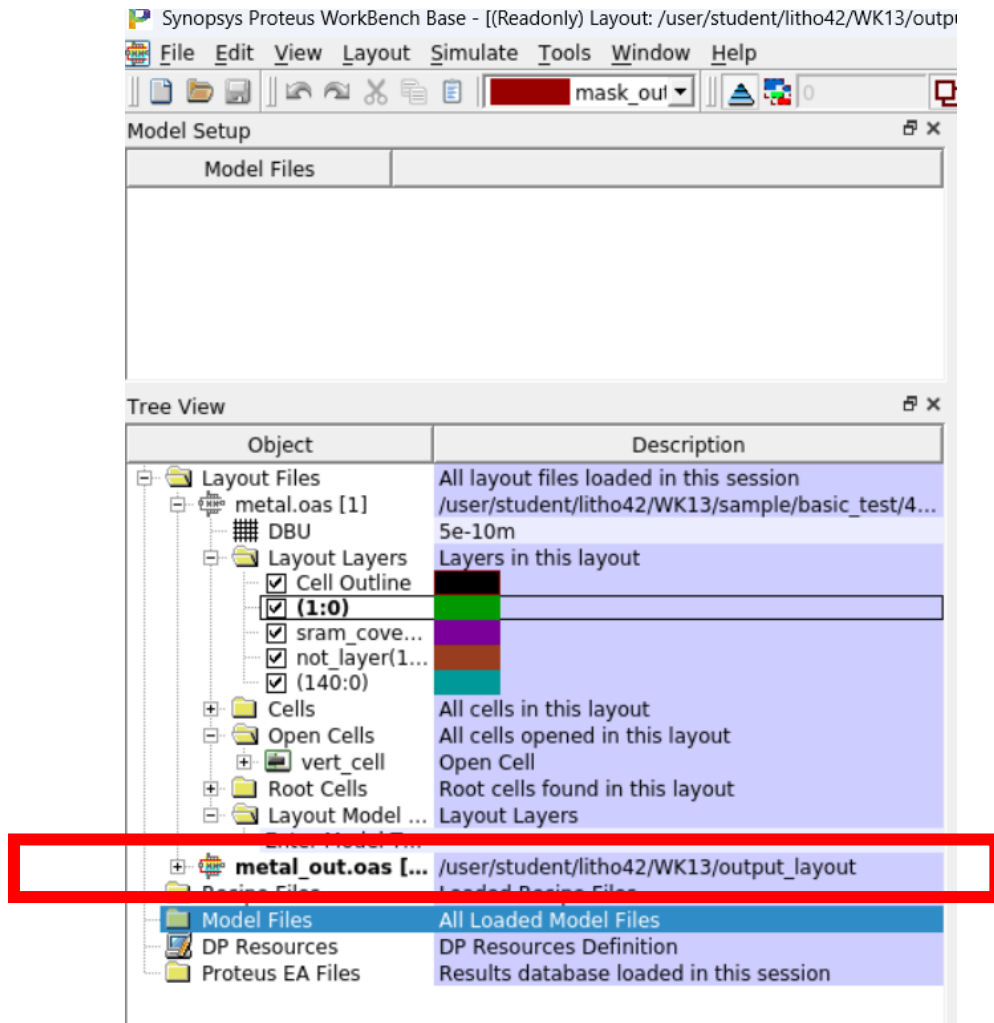
- Hierman : OPC하기 전 입력 Layout을 분석하여 Template 생성

- Corexec : 생성된 Template를 읽어서 위에서 적용한 Recipe Parameter 적용하여 OPC 수행

OPC 결과물 생성

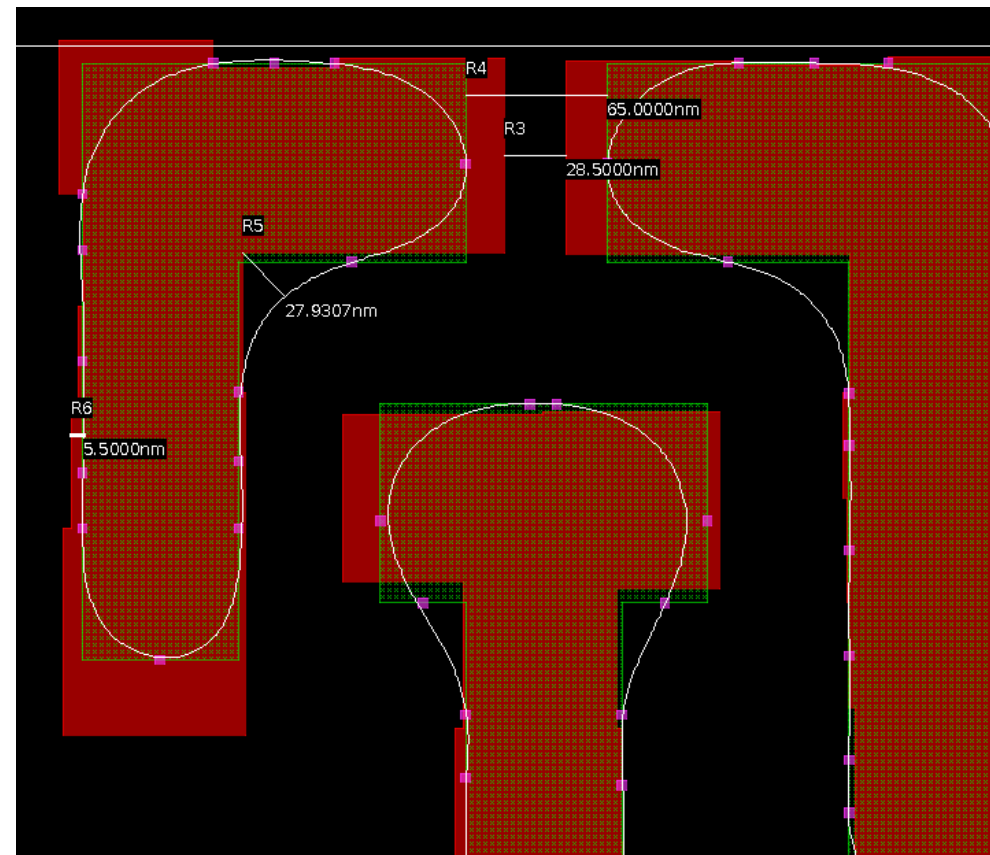
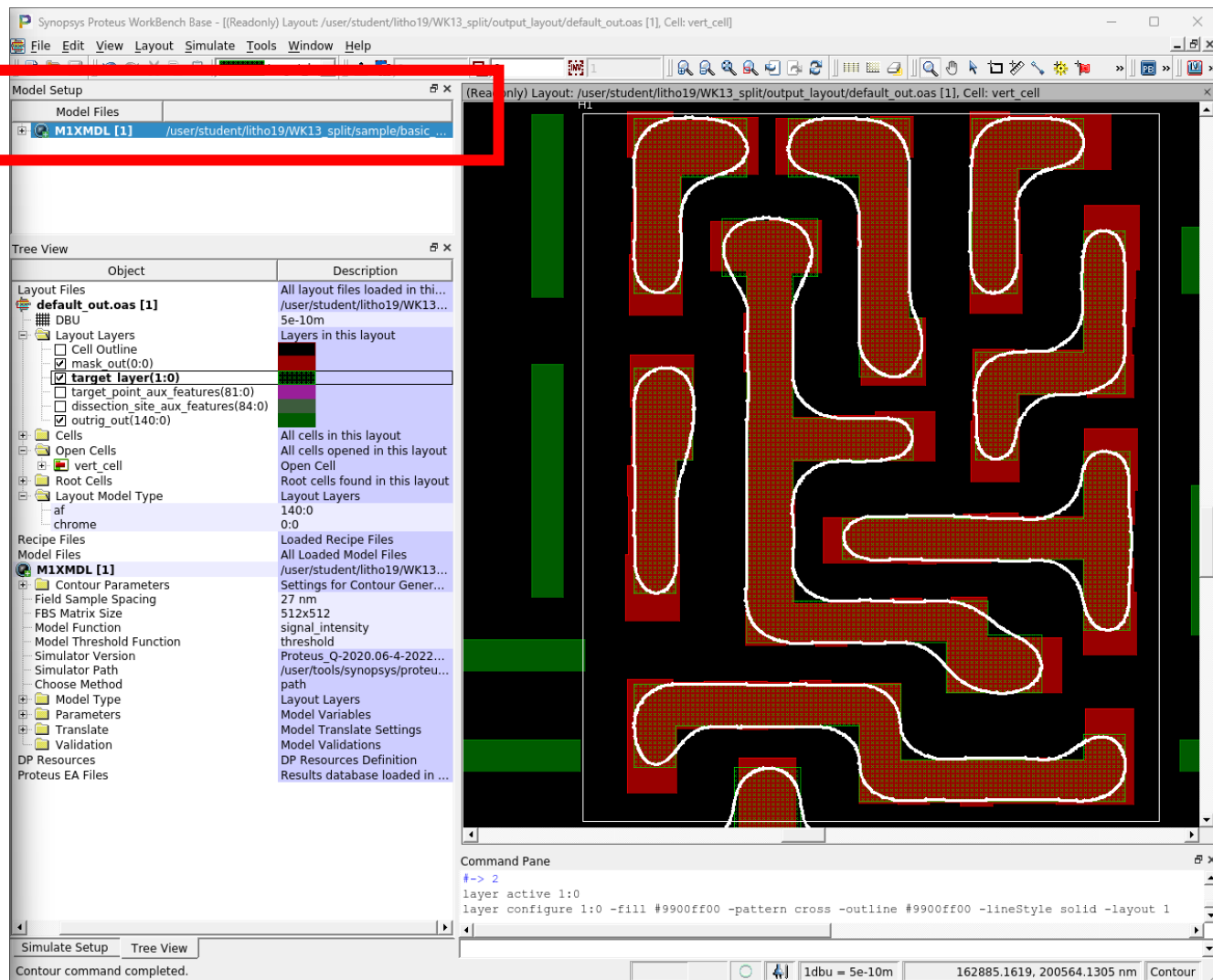
```
[litho42@ssudisu4 ~/WK13]$ cd output_layout  
[litho42@ssudisu4 output_layout]$ ls  
metal_out.oas  metal_out.oas.TINF
```

default: OPC 후 생성된 layout

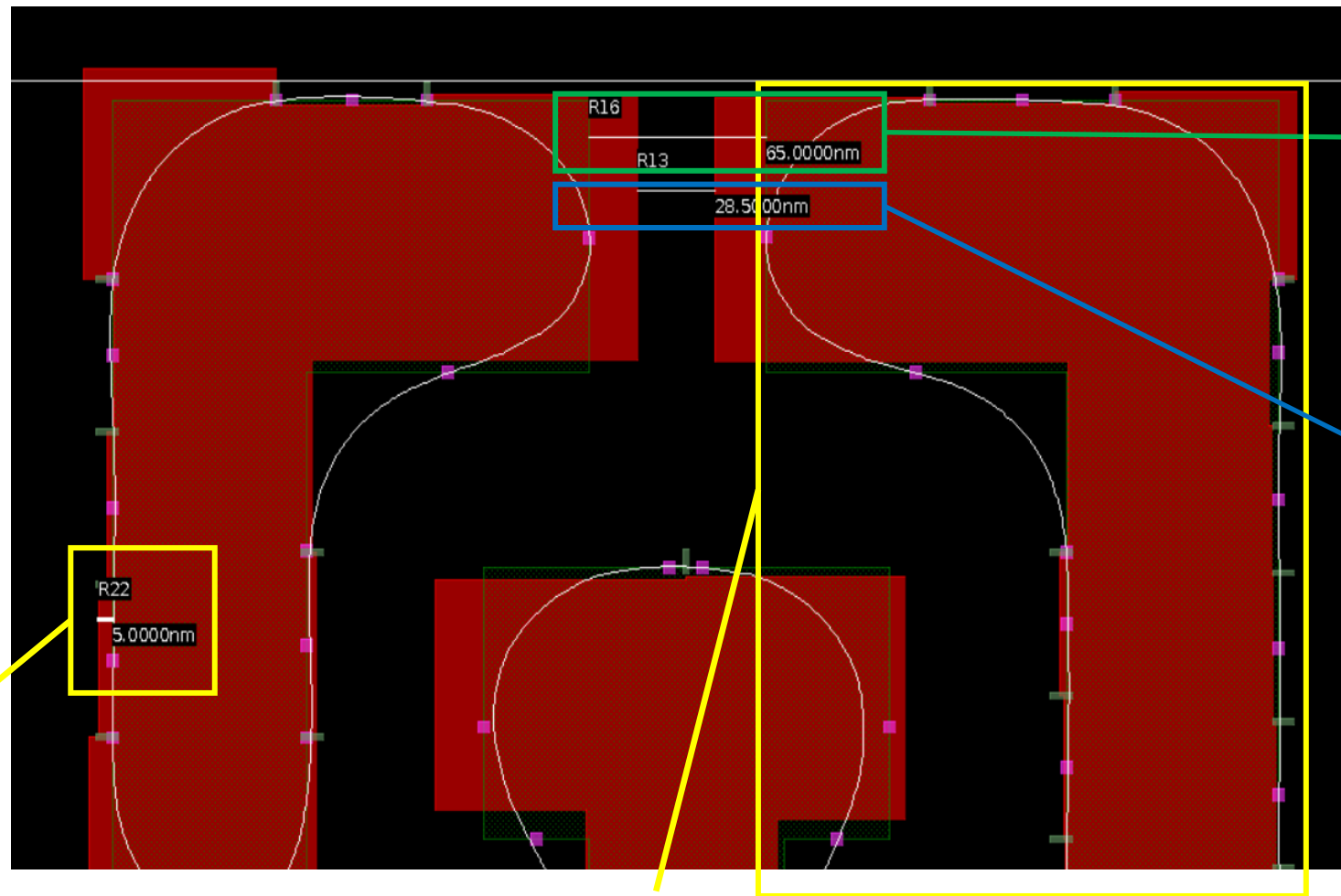


default: OPC 후 생성된 layout

- Model Set up / simulation 후 CD 측정



default: OPC 전 후 패턴 CD 측정



Target layout 기준
spacing : 65 nm

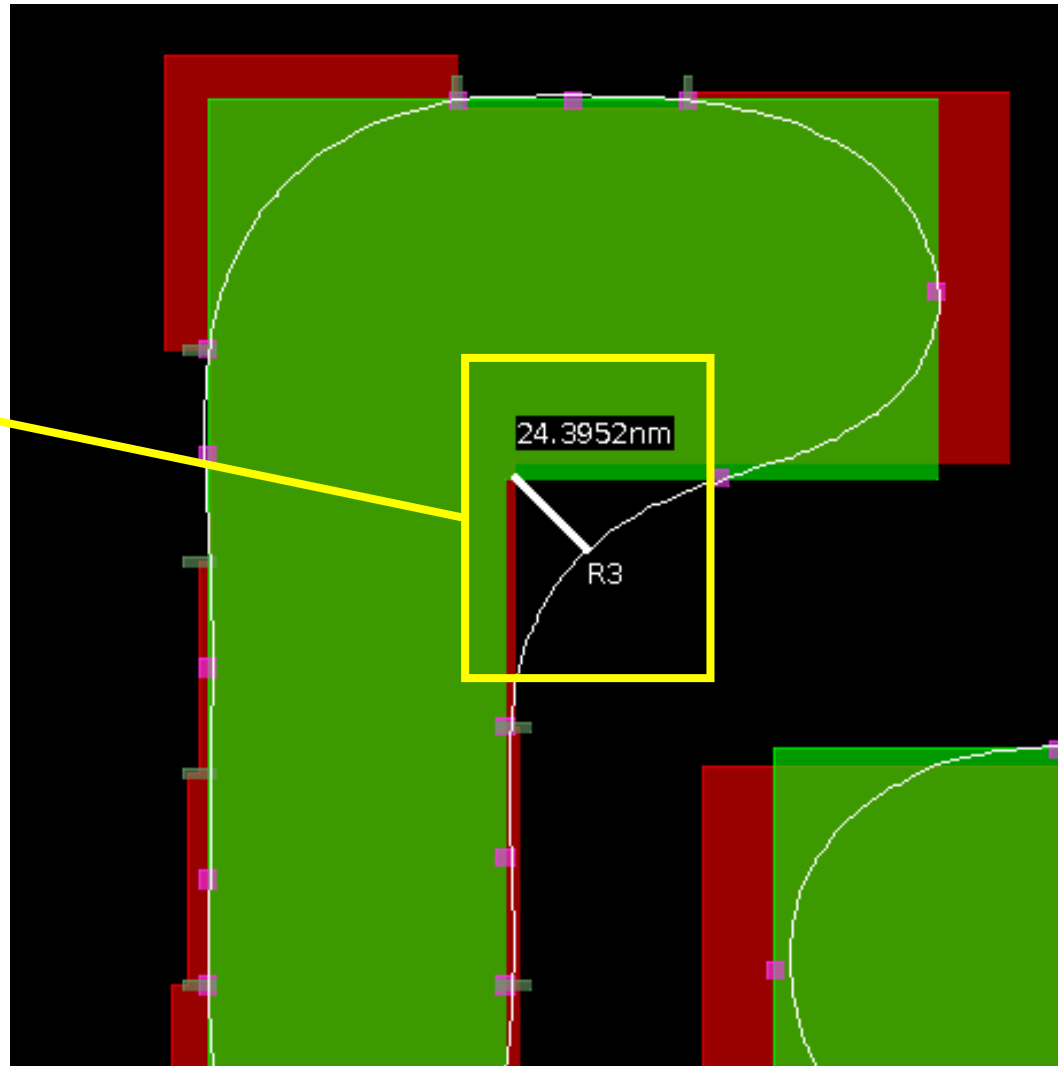
OPC MASK 기준
spacing = 28.5nm

OPC Mask patter과
Contour 간 CD차이
: 5nm

Contour
: 실제 웨이퍼에 인쇄될 것이라 예측되는 패턴

default: OPC 전 후 패턴 CD 측정

Corner Rounding
:24.39nm



Split: recipe 변경

	default	split	}	Feature Classification
<u>min_feature</u>	65	50		
	default	split	}	Dissection
<u>Dissection.out_vert_len</u>	60	50		
<u>Dissection.in_vert_len</u>	60	50		
<u>Dissection.nominal_segment_len</u>	50	40		
<u>Dissection.min_seg_len</u>	40	35		
<u>Dissection.num_run_segs</u>	5	6		
	default	split	}	Retarget
<u>Retargeting.vert_control</u>	1.0	0.8		

Feature Classification

Feature Classification

Design Rule:

<min_feature> 50

<jog_height> 10

Short Edges:

<min_correction_length> [(0.5*<min_feature>)]

Line Ends:

<end_width> [(2*<min_feature>)]

<min_end_extension> <min_correction_length>

- **Min_feature** : layout의 최소 CD(Critical Dimension)

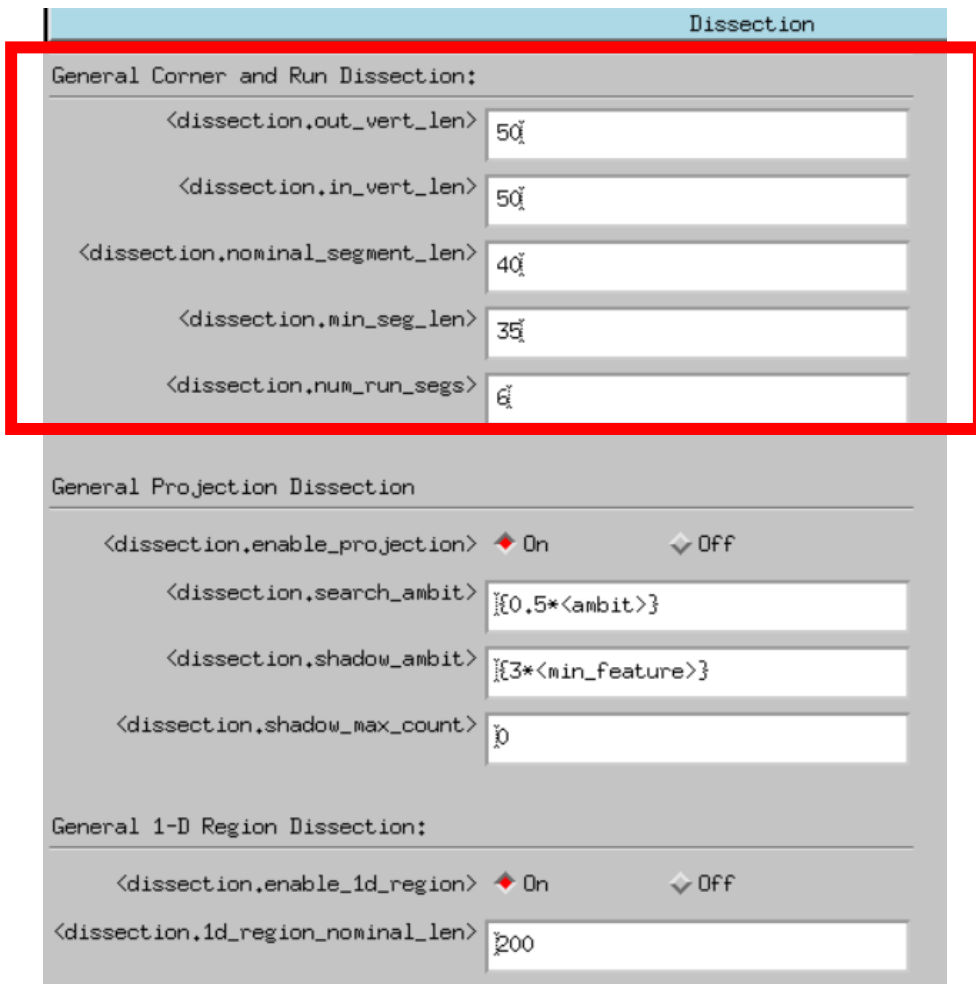
기존 default : 65nm → split : 50nm로 변경

예상 결과

: 더 작은 Feature를 인식하여 세밀한 OPC 보정

→ Corner 및 미세 패턴 영역에서 작은 패턴까지 고려할 수 있기 때문에 보정의 정확도가 향상 될 것이라 예상

Dissection



The screenshot shows a software configuration window titled "Dissection". A red rectangular box highlights the "General Corner and Run Dissection" section. Below this section are three other sections: "General Projection Dissection" and "General 1-D Region Dissection".

General Corner and Run Dissection:

- <dissection.out_vert_len> 50
- <dissection.in_vert_len> 50
- <dissection.nominal_segment_len> 40
- <dissection.min_seg_len> 35
- <dissection.num_run_segs> 6

General Projection Dissection:

- <dissection.enable_projection> ☒ On ☐ Off
- <dissection.search_ambit> $\{[0, 5 * \text{ambit}]\}$
- <dissection.shadow_ambit> $\{[3 * \text{min_feature}]\}$
- <dissection.shadow_max_count> 0

General 1-D Region Dissection:

- <dissection.enable_1d_region> ☒ On ☐ Off
- <dissection.1d_region_nominal_len> 200

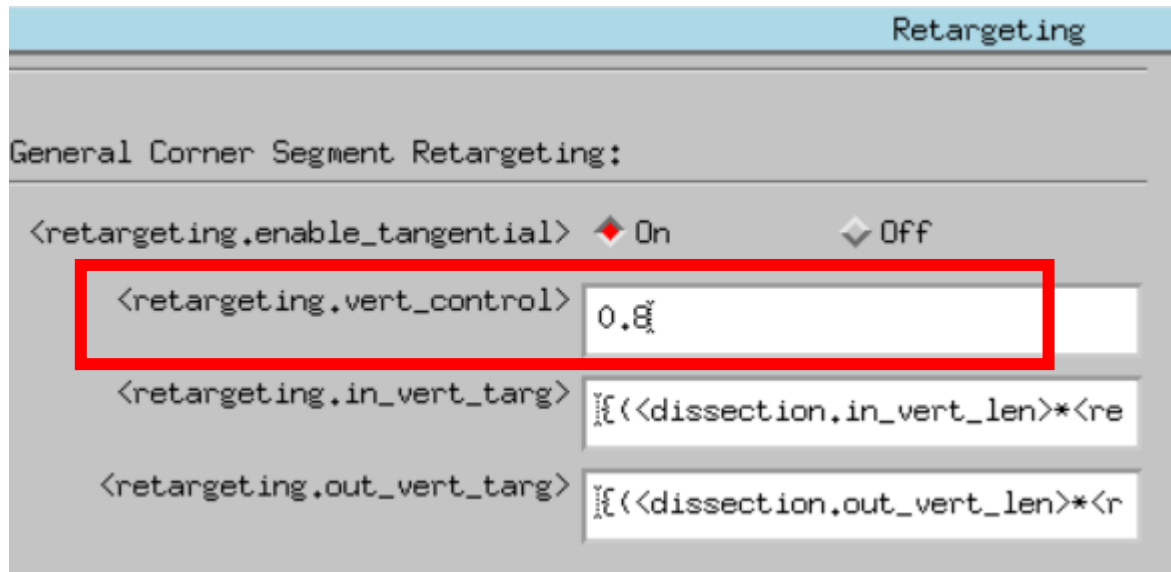
- Dissection
 - out_vert_len 60 → 50
 - in_vert_len 60 → 50
 - nominal_seg_len 50 → 40
 - min_seg_len 40 → 35
 - num_run_segs 5 → 6

예상 결과

: Edge를 더 세밀하게 분할하여 정밀한 OPC 보정 수행

→ Segment 수 증가를 통해 Corner 및 Edge 영역의 보정 정밀도 향상 될 것이라 예상

Retargeting



- **Retargeting.vert_control**
기존 default 1.0 → split 0.8로 감소

예상 결과

: Corner 부근의 보정 강도를 완화하여 과도한 보정 방지

→ Corner Rounding 보정 시 과도한 Edge 이동 감소 예상

Hierman setting / Corexec (Split)

```
[litho42@ssudisu4 ~/WK13_split]$ hierman split.pjx
```

hierman

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```
[litho42@ssudisu4 ~/WK13_split]$ corexec split.pjx
```

corexec

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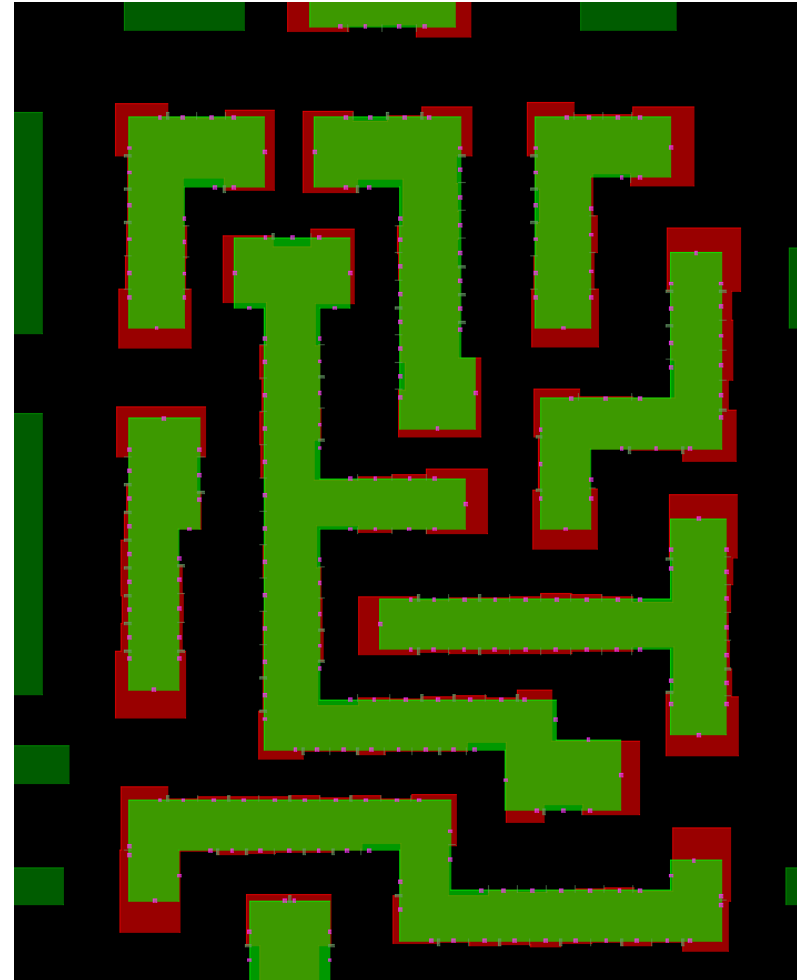
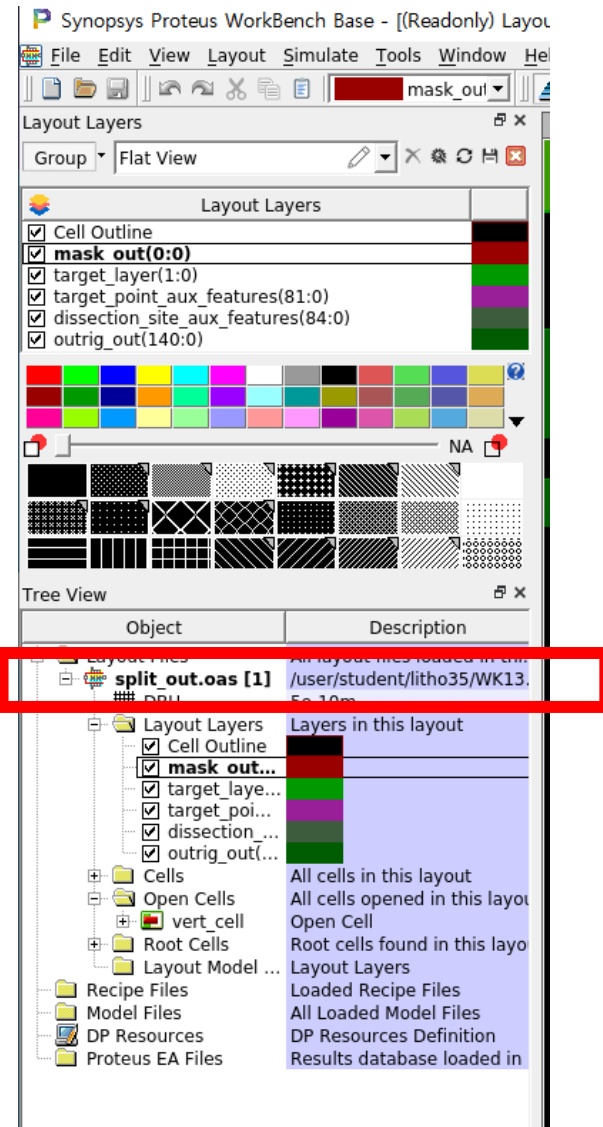
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- Hierman : OPC하기 전 입력 Layout을 분석하여 Template 생성

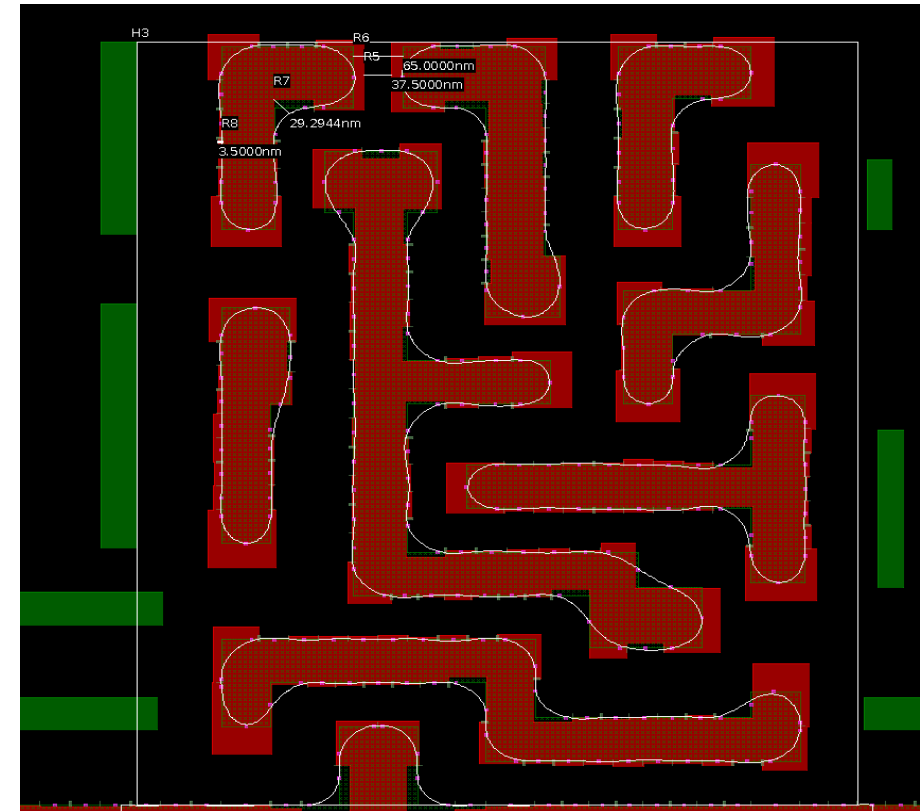
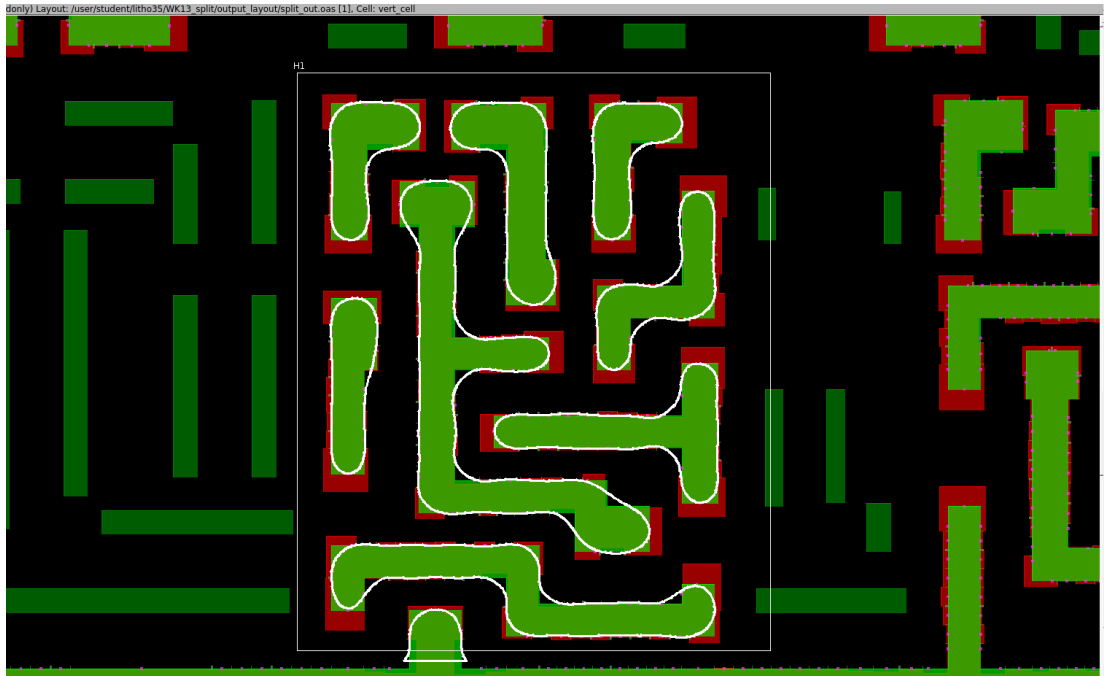
- Corexec : 생성된 Template를 읽어서 위에서 적용한 Recipe Parameter 적용하여 OPC 수행

split: OPC 후 생성된 layout

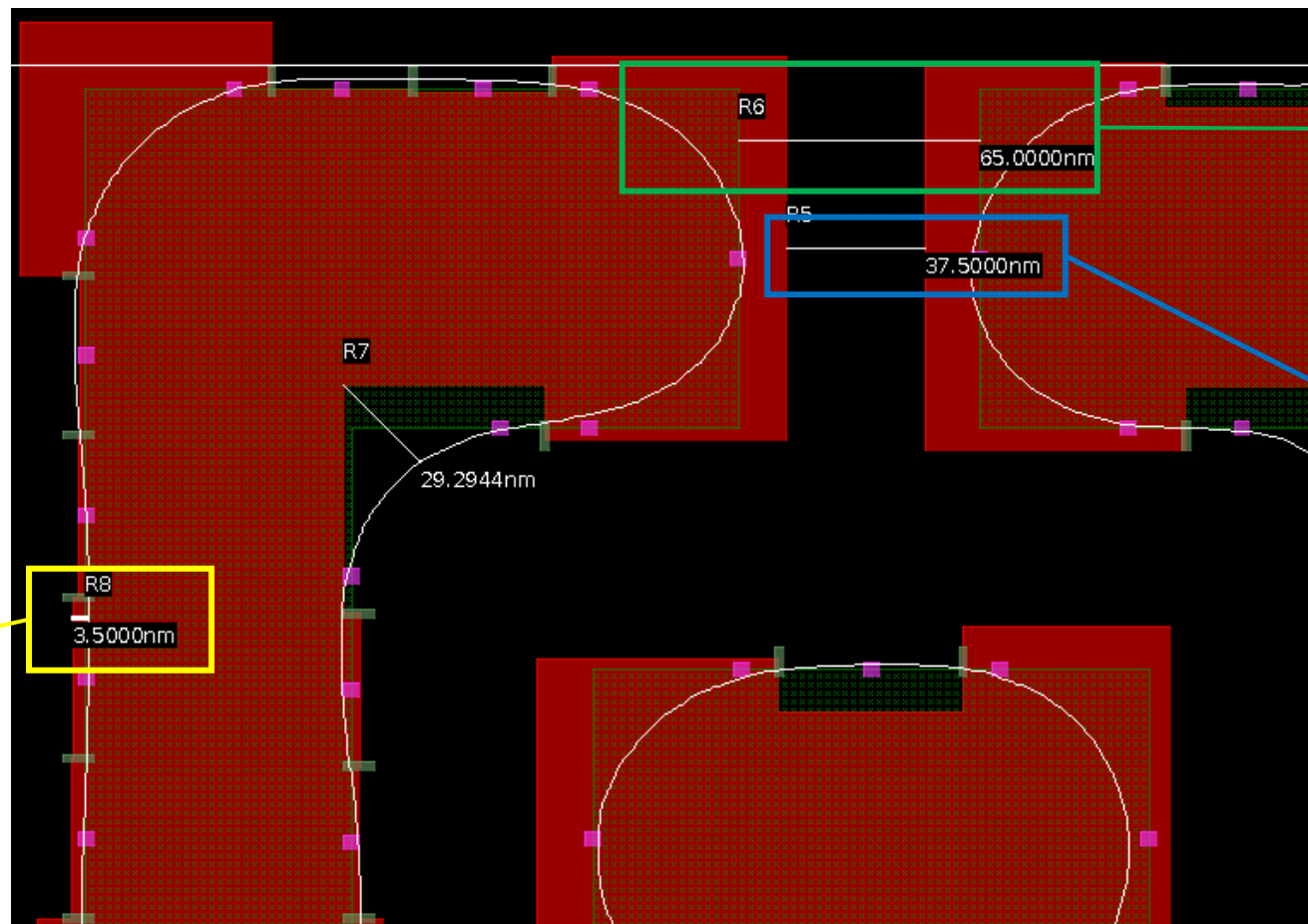


split: OPC 후 생성된 layout

- simulation 후 CD 측정



split: OPC 전 후 패턴 CD 측정



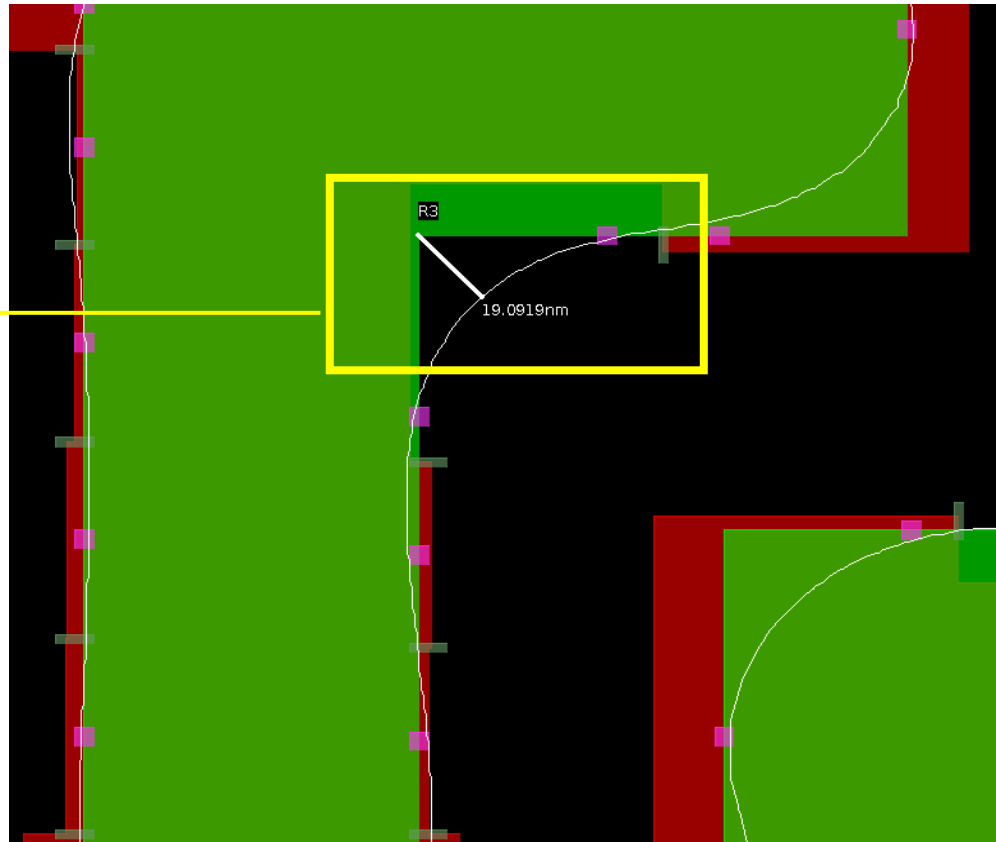
Target layout 기준
spacing : 65 nm
- default와 당연히 같음

OPC MASK 기준
spacing = 37.5nm
(default : 28.5nm)

OPC Mask patter과
Contour 간 CD차이
: 3.5nm
(default : 5nm)

split: OPC 전 후 패턴 CD 측정

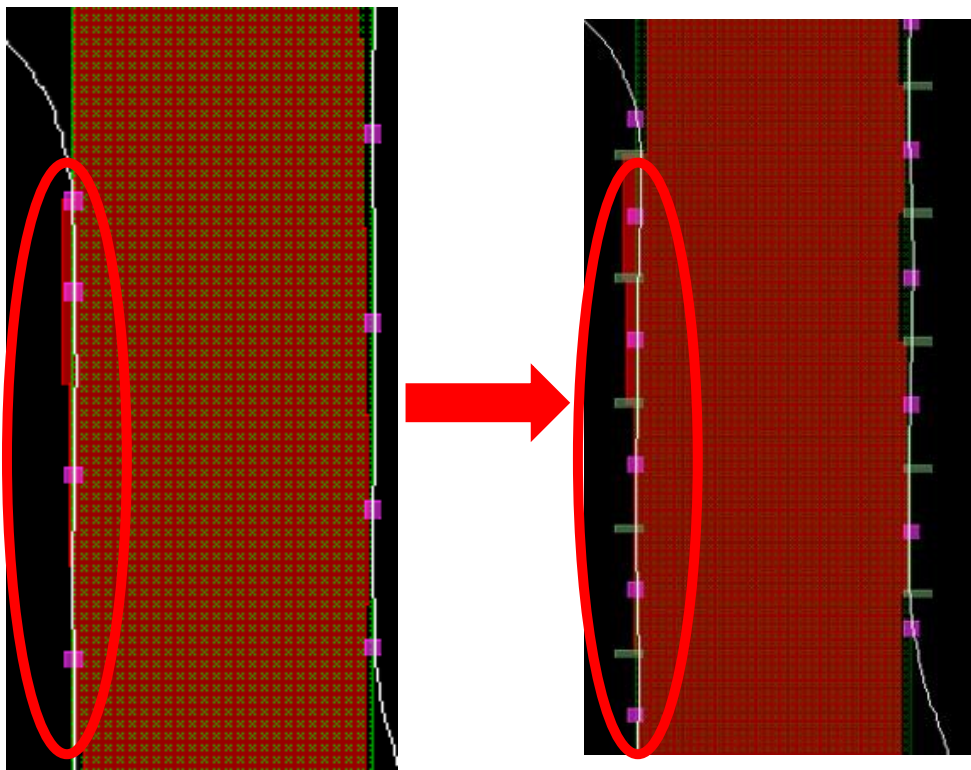
Corner rounding
: 19.1nm
(default : 24.39nm)



OPC MASK 기준
spacing = 37.5nm
(default : 28.5nm)

Default vs Split

1. Dissection segment point 증가



Dissection.nominal_segment_len: 50 → 40

Dissection.num_run_segs: 5 → 6

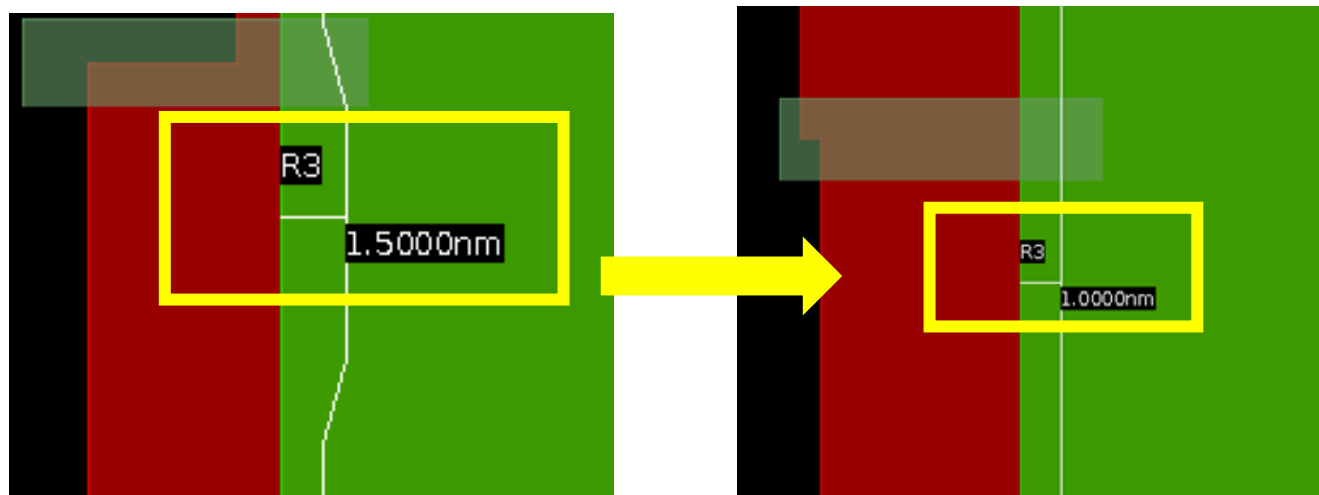
Dissection.min_seg_len: 40 → 35

Dissection.out/in_vert_len: 65 → 50

Contour segment 세분화가 진행되었다.

Default vs Split

2. Contour narrowing 감소



Dissection.nominal_segment_len: 50 → 40

Dissection.num_run_segs: 5 → 6

Retargeting.vert_control: 1.0 → 0.8

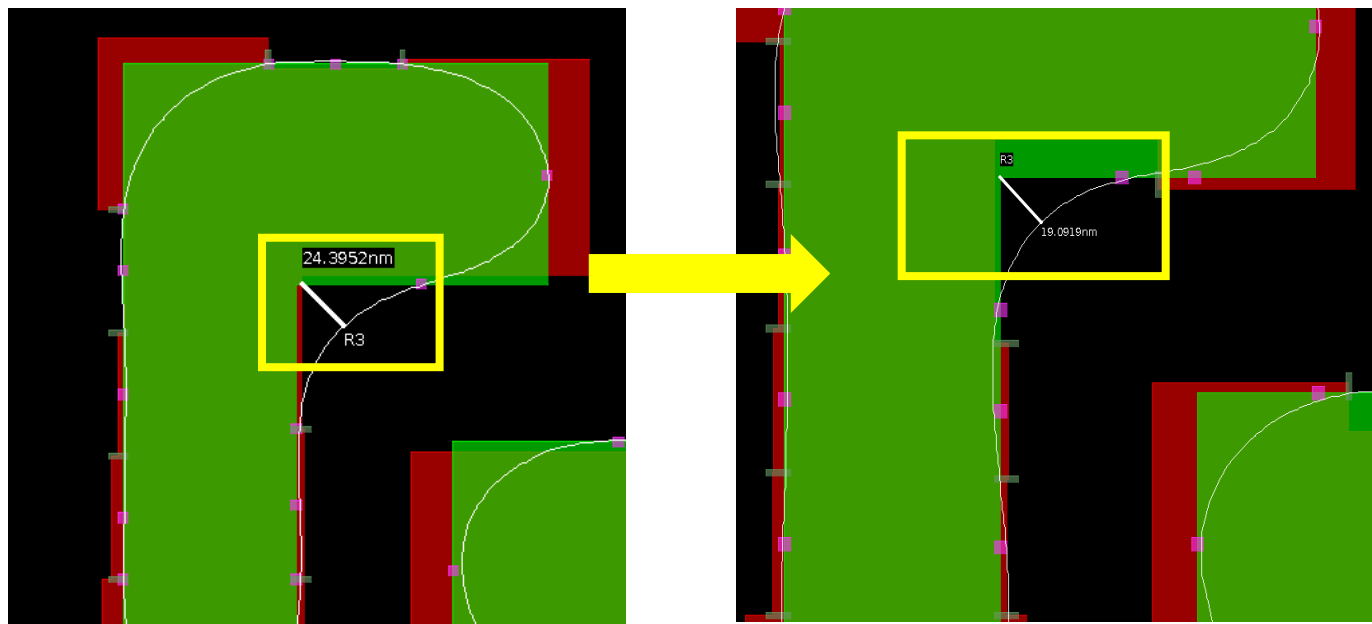
Default : MASK layout – Contour 거리 : 1.5nm

Split : MASK layout – Contour 거리: 1nm

➔ Contour narrowing 감소

Default vs Split

3. Corner rounding 개선



Dissection.out/in_vert_len: 65 → 50
Dissection.nominal_segment_len: 50 → 40
Dissection.num_run_segs: 5 → 6

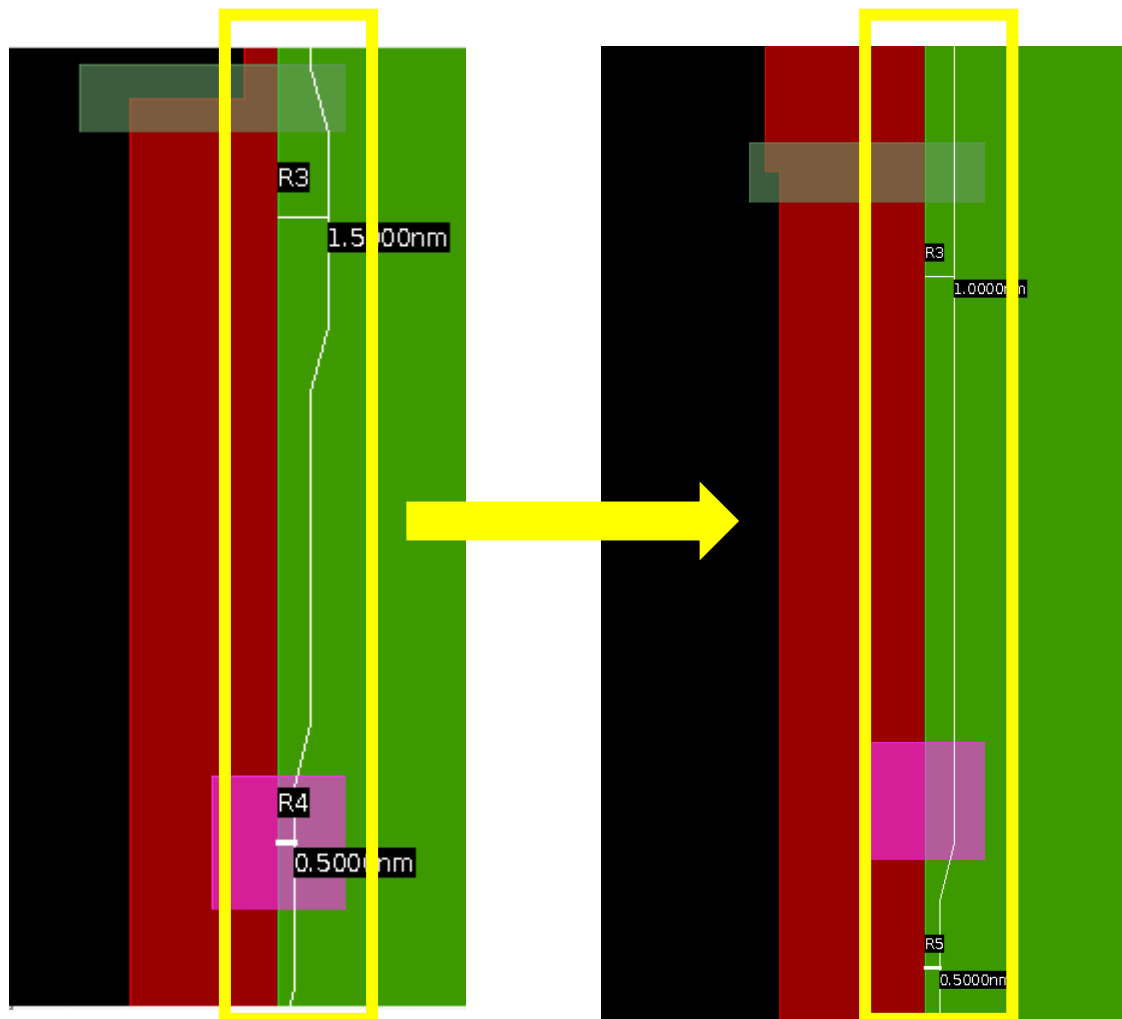
Corner 영역 segmentation

Default Corner Rounding 정도 : 24.4nm
Split Corner Rounding 정도 : 19.1nm

➔ Corner rounding 개선

Default vs Split

4. Line edge roughness 감소



Dissection.nominal_segment_len: 50 → 40

Dissection.num_run_segs: 5 → 6

Retargeting.vert_control: 1.0 → 0.8

육안으로 보아도 **Line Edge Roughness(LER)** 감소

Conclusion

1. Split 조건 적용을 통해 contour segmentation 세분화

- Smaller feature correction 가능
- Dissection segment point 증가
- Edge contour fitting 정밀도 향상

2. Pattern quality 개선 확인

- Contour narrowing 감소
- Contour rounding 개선
- Line Edge Roughness 감소

Conclusion

Split parameter optimization을 통해 pattern quality 향상

미세 pattern에서 안정적인 contour control 가능성 확인